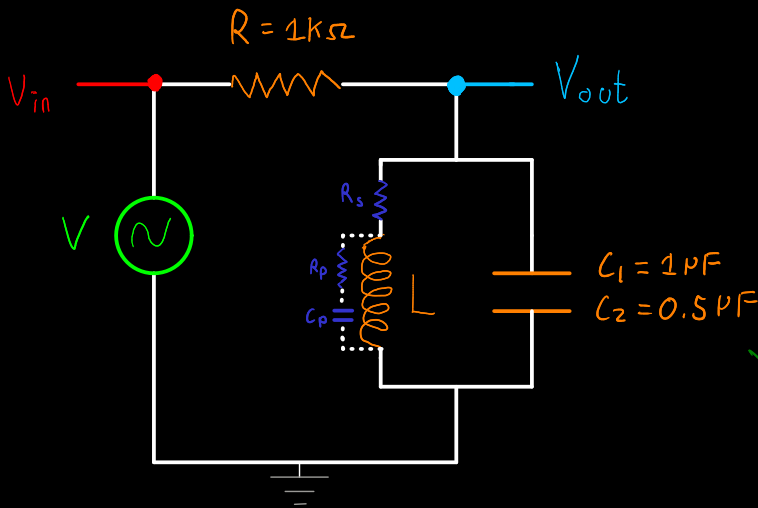
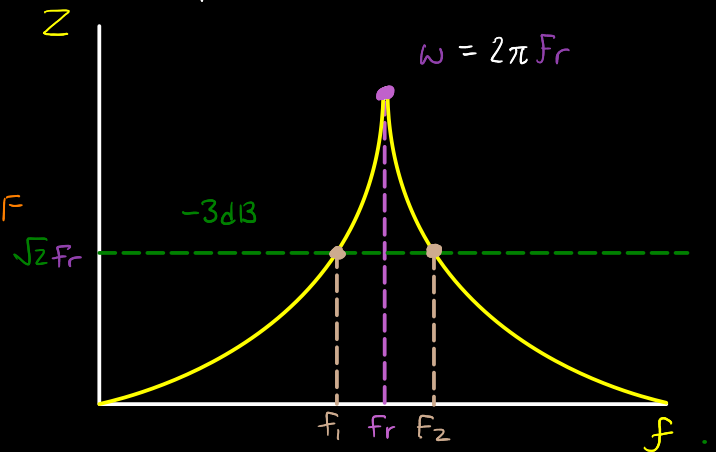


- R_s : Series Resistance
- R_p : Parallel Resistance
- C_p : Parallel Capacitance



- For a range of frequency Measure the Impedance : $Z = \frac{V_{out}}{V_{in}}$

- We expect



- Voltage Divider : $V_{out} = \frac{(R_s)(V_{in})}{R + R_s}$

$$\rightarrow Z = \frac{V_{out}}{V_{in}} = \frac{R_s}{R + R_s}$$

(ESR)

- Equivalent Series Resistance : $R_s = \frac{Z R}{1 - Z}$

(EPC)

- Equivalent Parallel Capacitance :

- Find f_r for C_1 and C_2

$$\Rightarrow f_{01} = \frac{1}{2\pi\sqrt{L[C_1 + C_p]}} \quad \Rightarrow f_{02} = \frac{1}{2\pi\sqrt{L[C_2 + C_p]}}$$

$$\rightarrow \frac{f_{02}}{f_{01}} = \frac{2\pi\sqrt{L[C_1 + C_p]}}{2\pi\sqrt{L[C_2 + C_p]}} \quad \rightarrow \frac{f_{02}}{f_{01}} = \sqrt{\frac{C_1 + C_p}{C_2 + C_p}}$$

$$\rightarrow \left(\frac{f_{02}}{f_{01}}\right)^2 = \frac{C_1 + C_p}{C_2 + C_p} \quad \rightarrow C_2 \left(\frac{f_{02}}{f_{01}}\right)^2 + C_p \left(\frac{f_{02}}{f_{01}}\right)^2 - C_p = C_1$$

$$\rightarrow C_p = \frac{C_1 - C_2 \left(\frac{f_{02}}{f_{01}}\right)^2}{\left(\frac{f_{02}}{f_{01}}\right)^2 - 1}$$

(EPR)

- Equivalent parallel Resistance :

$$Z_L = R_s + \frac{i\omega L \cdot R_p}{i\omega L + (1 - \omega^2 L[C_2 + C_p]) \cdot R_p}$$

- Calculate Inductance : $L = \frac{1}{\omega^2 [C_2 + C_p]}$

- I Find that $L = \frac{1}{\omega^2 C_2}$
Gives a close enough answer.

- Calculate Q_L of Inductor : $Q_L = \frac{\omega L}{R_s}$

- Quality Factor : $Q = \frac{f_r}{f_2 - f_1}$

$$Q_L = \frac{X_L}{R_s}$$

- Resonance Frequency : $f_r = \frac{1}{2\pi\sqrt{L[C_1 + C_p]}}$

- Impedance of Inductor : $Z_L^2 = R_{oc}^2 + X_L^2$

- Inductor Reactance : $X_L = \omega L$